

The storms are dangerous. But the risk story is changing.

U.S. fatalities, population exposure, and the economics of adaptation | 1940–2025

240

DEATHS / YEAR

Average tornado + flood + hurricane fatalities across 1940–2025.

–68%

DEATH INTENSITY

From ~1.88 to ~0.61 deaths per million people per year, 1940s to 2010s.

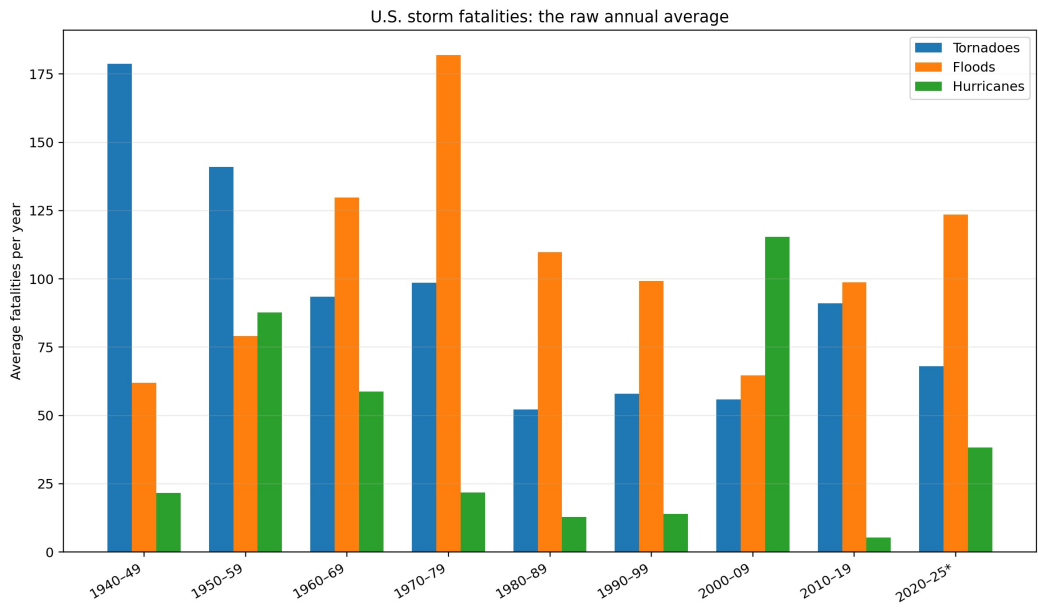
382

HEAT DEATHS / YEAR

Average in 2020–25; heat changes how the modern record looks.

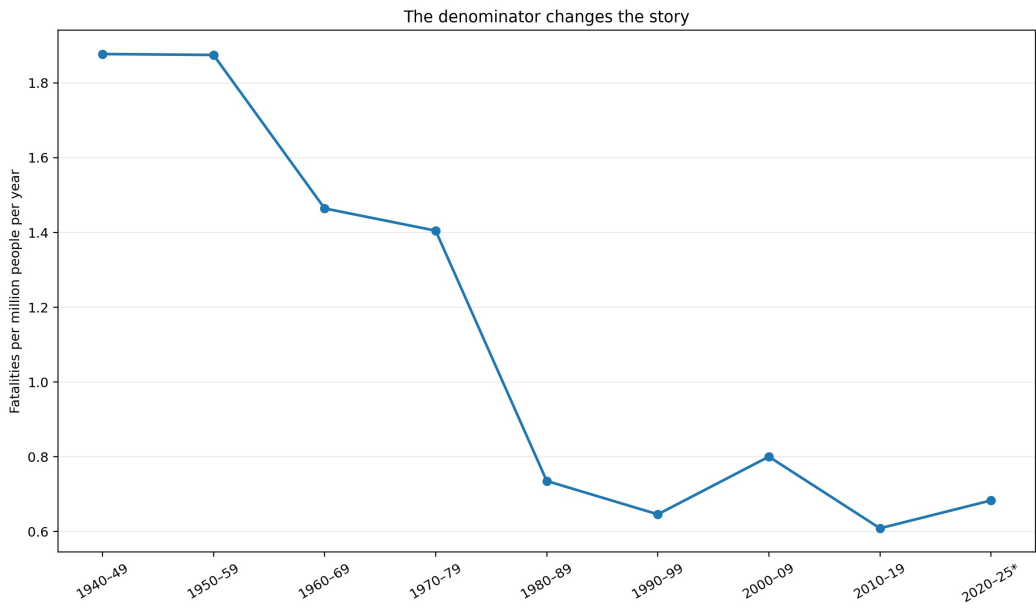
1 Raw fatalities fall unevenly

The number of deaths per year is not a smooth trend—and individual disasters can dominate a decade.



2 The denominator matters

Population growth makes raw counts a poor measure of per-person risk.

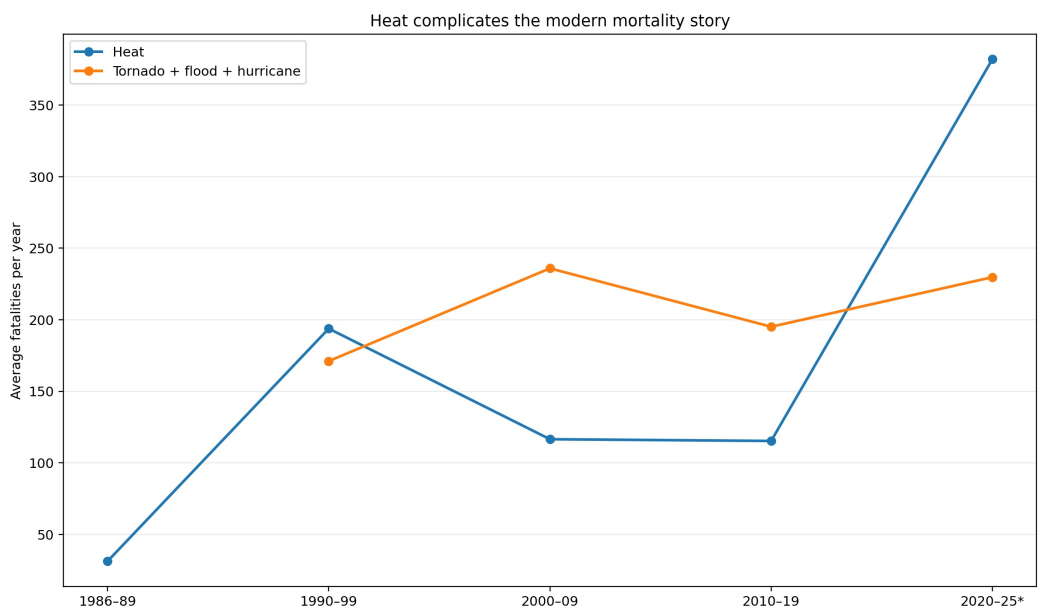


What the data suggest

The raw record is consistent with substantial improvement in protection from tornadoes, floods, and hurricanes: the three-hazard average was about 262 deaths per year in the 1940s and about 195 in the 2010s. The population denominator makes the change much clearer. The U.S. became far more populous while the death rate from these three hazards fell. That pattern is consistent with adaptation—forecasting, warning systems, construction, evacuation, emergency response, and other technologies—but the data alone cannot identify which changes caused the decline.

3 Heat complicates the story

NWS heat-fatality reporting begins in 1986; 2020–25 is only six years.

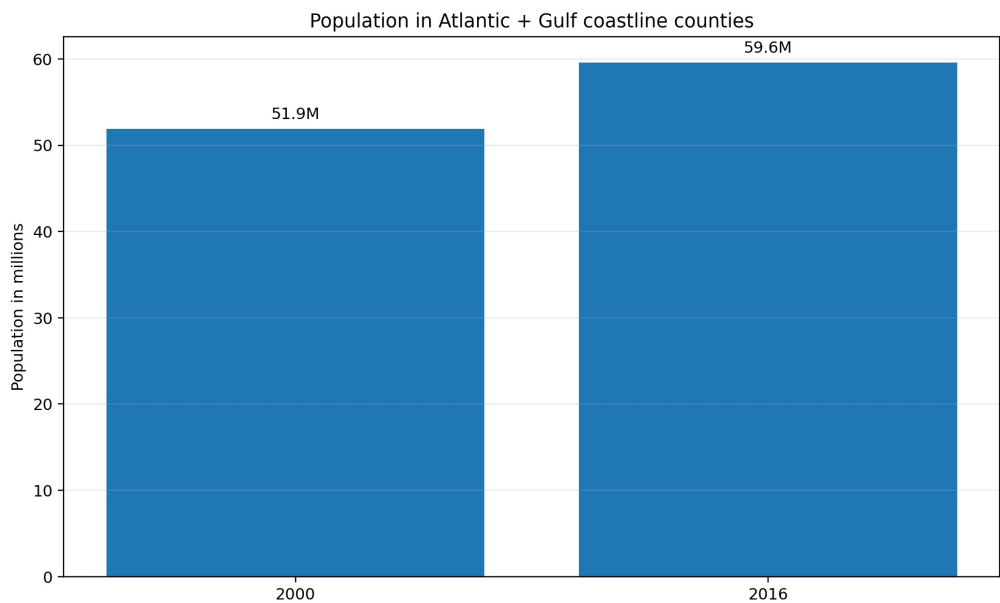


A crucial measurement caveat

The NWS record does not show a simple all-weather decline. Heat fatalities averaged roughly 115 per year in the 2010s and 382 per year in 2020–25, with a huge 1995 spike. NWS counts deaths attributable to a weather hazard in Storm Data. Burke et al. (2025) instead estimate mortality attributable to ambient temperature using detailed mortality and temperature data. Their estimate is not directly comparable to the NWS count.

4 Adaptation is not the same thing as retreat

Atlantic + Gulf coastline counties grew from 51.9M people in 2000 to 59.6M in 2016.



The economics of risk

People do not choose locations based only on hazard exposure. Jobs, housing, amenities, family networks and infrastructure create benefits from living in particular places. A risky location can remain attractive when its expected benefits exceed perceived costs. Public institutions meanwhile invest in forecasts, warnings, building standards, infrastructure and emergency response. The economic question is therefore not simply “How do we stop the hazard?” but “Which exposures are worth reducing, at what cost, and for whom?”

But wait.

A falling fatality rate is evidence of an outcome, not proof that climate risk is falling. These data do not isolate climate change, exposure changes, technology, income, forecasting, or emergency management. Nor do they measure all temperature-related mortality. Burke et al. find that cold accounts for more temperature-attributable mortality than heat in their sample, while hotter places show some adaptation to local climate. The safest conclusion is narrower: the U.S. has become much less deadly, per person, from the three storm hazards counted here—while the broader temperature-mortality story remains more complicated.